TCV-ϕ V: cohesive quantum microstructure from CC-Φ_1 and a controlled bridge to CC-Φ_2 flavour templates

Cyrille LECROQ[1]

[1] *Independent Researcher*

(Dated: March 26, 2026)

This fifth paper develops the quantum-microstructure layer of the canonical TCV-ϕ programme. Our scope is to formulate a reproducible CC-Φ_1 cluster framework, quantify its curvature response, and connect it to coarse-grained effective quantities used in previous papers. For the flavour bridge, the historical twisted-torus benchmark is retained as the geometry that first exposed the relevance of non-trivial topology, while an extended scan within the same topological class now identifies twisted-multi-ring as the strongest current exploratory candidate. No complete theory-level fit is claimed. All claims are restricted to script-backed diagnostics archived in the Paper V workspace.

I. INTRODUCTION

Paper V follows Papers I–IV and keeps the same canonical EFT field content and notation. The focus is the microscopic cohesive layer (CC-Φ_1) and its controlled link to exploratory CC-Φ_2 flavour templates. In this revision, the flavour subsection is organized around geometry selection: we first compare several cluster topologies with the same PMNS-oriented criteria, then refine the leading topological family and promote the strongest resulting candidate to the primary exploratory benchmark.

II. CANONICAL EFT BASELINE

We keep exactly the same action and parameter hierarchy as in Papers I–IV. No new fundamental fields or couplings are introduced in this paper.

III. LOCAL CC-Φ_1 CLUSTERS: SETUP

We adopt a minimal cluster quantization toy consistent with the CC-Φ_1 preprint construction of Ref. [**?**]:

$$L_c(R) \sim \frac{1}{m_{\rm eff}(R)}, \qquad m_{\rm eff}^2(R) = m_1^2 + \xi_0 R, \tag{1}$$

and

$$\Xi_n(R) = m_{\rm eff}^2(R) + \left(\frac{n\pi}{L_c(R)}\right)^2. \tag{2}$$

Here Ξ_n has mass-squared units, while the corresponding mode energy scale is

$$\omega_n(R) = \sqrt{\Xi_n(R)}. \tag{3}$$

The baseline benchmark used by `compute_ccphi1_spectrum.py` is $m_1 = 10^{-8}$, $\xi_0 = 0.1$, and $n_{\rm modes} = 3$.

IV. CC-Φ_1 SPECTRUM AND CURVATURE RESPONSE

From `ccphi1_spectrum.json`, the baseline values at $R = 0$ are

$$\Xi_1 \simeq 1.09 \times 10^{-15}, \qquad \Xi_2 - \Xi_1 \simeq 2.96 \times 10^{-15}, \tag{4}$$

which corresponds to $\omega_1(0) \simeq 3.30 \times 10^{-8}$ and coherence length $L_c(0) \simeq 10^8$ in Planck units. The curvature scan (`compute_ccphi1_curvature_response.py`) uses $0 \le R \le 2 \times 10^{-15}$ and yields

$$\frac{\Xi_1(R)}{\Xi_1(0)} \in [1, 3], \qquad \frac{L_c(R)}{L_c(0)} \in [0.577, 1]. \tag{5}$$

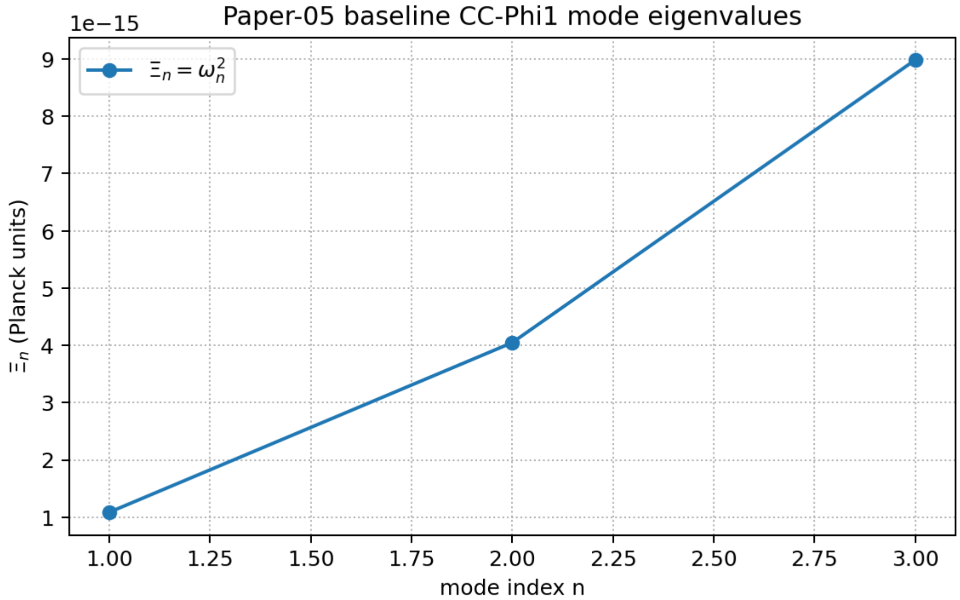

FIG. 1. Baseline CC-Φ_1 mode spectrum from `compute_ccphi1_spectrum.py`.

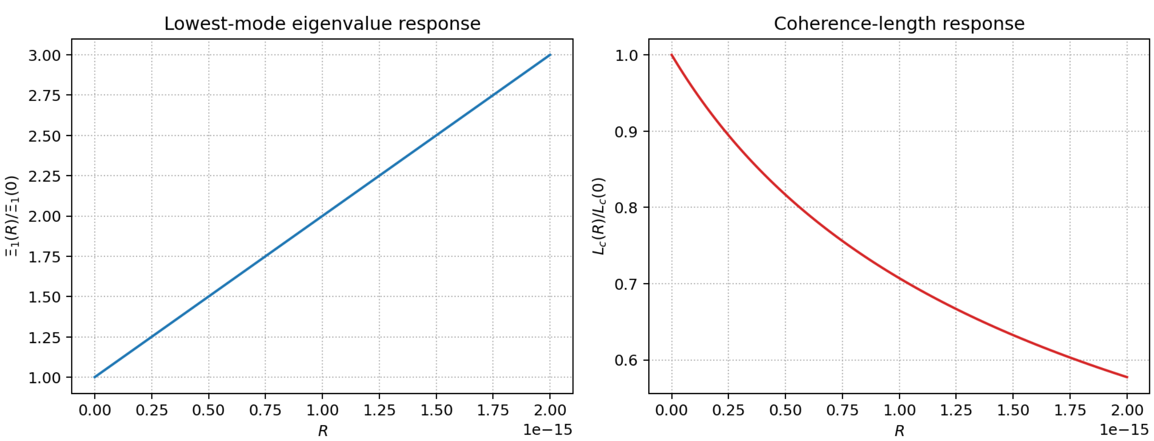

FIG. 2. Curvature response of the lowest mode and coherence length from `compute_ccphi1_curvature_response.py`.

V. COARSE-GRAINING AND EFFECTIVE MACROSCOPIC CONSISTENCY

At the current toy level we use a proxy density $n_{\mathrm{CC}} \sim m_1^3$ and define

$$\rho_{\mathrm{cc,eff}} \sim n_{\mathrm{CC}}\,\omega_1, \qquad p_{\mathrm{proxy}} = w_{\mathrm{proxy}}\rho_{\mathrm{cc,eff}}, \tag{6}$$

with the illustrative closure choice $w_{\mathrm{proxy}} = -1/2$ in the current script benchmark. This w_{proxy} is a toy coarse-graining parameter and should not be interpreted as the final EFT vacuum equation of state. The script `compute_ccphi1_coarse_grain.py` gives

$$\rho_{\mathrm{cc,eff}} \in [3.30, 5.71] \times 10^{-32}, \tag{7}$$

over the same curvature window. These values are used as consistency diagnostics, not as a final microscopic QFT derivation.

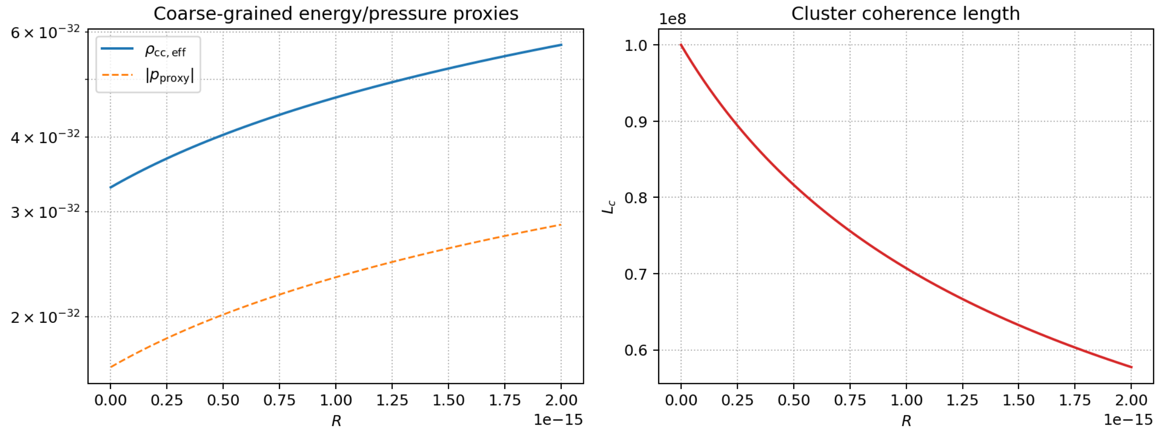

FIG. 3. Toy coarse-graining diagnostics from `compute_ccphi1_coarse_grain.py`.

VI. EXPLORATORY BRIDGE TO CC-Φ_2 FLAVOUR SECTOR: FROM TWISTED TORUS TO TWISTED MULTI RING

We keep the same conservative status as in previous drafts: this is an exploratory toy layer, not a precision global fit. The central update is geometric. Instead of using a single fixed cluster ansatz, we now distinguish between a historical and a current benchmark. The twisted-torus (Möbius-ladder) cluster remains the geometry that first produced a convincing PMNS-like signal under a common protocol. However, a dedicated follow-up scan within the same closed twisted-topology family points to a twisted-multi-ring cluster as the stronger current flavour candidate.

The effective flavour extraction remains in the same form,

$$M_\ell = \mathrm{diag}(m_e, m_\mu, m_\tau) + \delta M_\ell, \qquad M_\nu = \mathrm{diag}(m_{\nu_1}, m_{\nu_2}, m_{\nu_3}) + \delta M_\nu, \tag{8}$$

with

$$U_{\rm PMNS} = U_\ell^\dagger U_\nu. \tag{9}$$

The key difference is that δM_ℓ and δM_ν are no longer tied to a single legacy template. The initial topology-aware benchmark is generated from twisted-torus mode structure in `compute_twisted_torus_flavor_toy.py`, and the refined benchmark is generated from `compute_twisted_multi_ring_flavor_toy.py`. The integer parameter `twist_shift` controls the staggered cross-channel offset between neighbouring rings; the best current candidate uses `twist_shift` $= 1$.

For the best script-backed twisted-multi-ring benchmark we obtain

$$\theta_{12} \simeq 32.1^\circ, \quad \theta_{13} \simeq 7.76^\circ, \quad \theta_{23} \simeq 45.3^\circ, \tag{10}$$

with a common PMNS proximity score $S_{\rm PMNS} \simeq 0.24$ (lower is better in our normalization). The PMNS reference point used in this toy score is $\theta_{12}^{\rm ref} = 33.0^\circ$, $\theta_{13}^{\rm ref} = 8.6^\circ$, $\theta_{23}^{\rm ref} = 45.0^\circ$, consistent with current global-fit central values ([5]). In the same scan,

$$n_{\rm strict} = 164, \qquad n_{\rm relaxed} = 285, \tag{11}$$

for $n_{\rm natural} = 3949$ accepted points. For comparison, the historical twisted-torus benchmark gave

$$\theta_{12} \simeq 34.6^\circ, \quad \theta_{13} \simeq 8.29^\circ, \quad \theta_{23} \simeq 41.6^\circ, \tag{12}$$

with $S_{\rm PMNS} \simeq 0.52$ and 82/4223 strict points. The dedicated twisted-multi-ring follow-up therefore improves both the common PMNS score and the strict-yield fraction without changing the canonical EFT field content. The reactor angle θ_{13} lies below the current experimental central value in the best twisted-multi-ring candidate, but remains inside the strict window and is therefore consistent with the exploratory status of this benchmark. This remains an exploratory indication of geometric viability, not a final flavour solution.

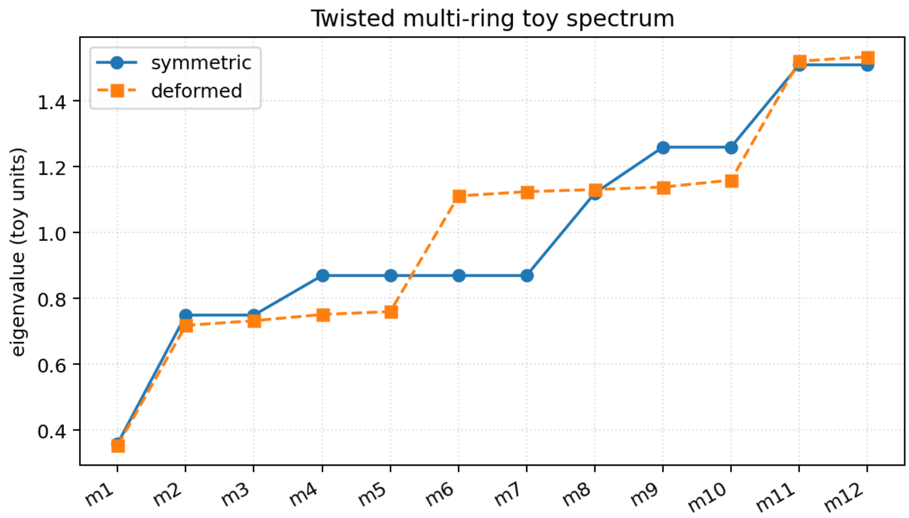

FIG. 4. Twisted-multi-ring flavour spectrum from `compute_twisted_multi_ring_flavor_toy.py`.

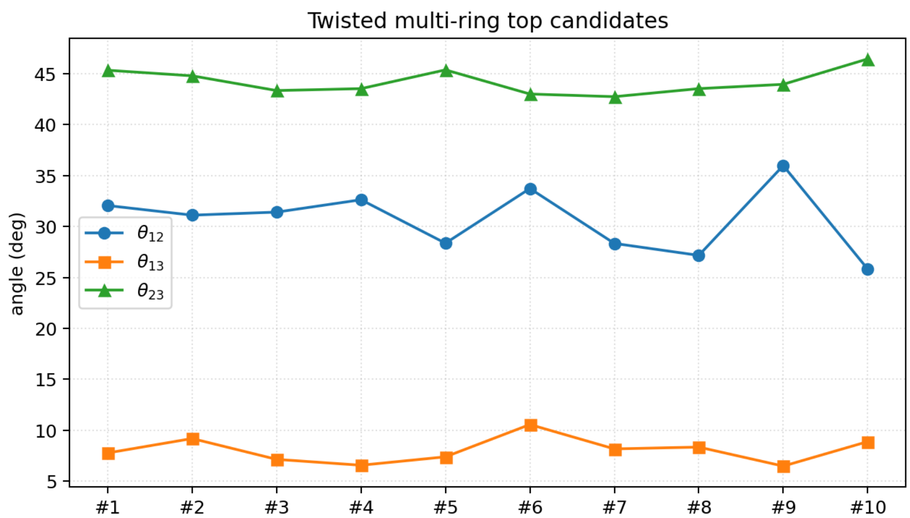

FIG. 5. Top twisted-multi-ring flavour candidates (script-backed).

VII. GEOMETRIC EXPLORATION AND MODEL SELECTION DIAGNOSTICS

To avoid hidden fine tuning, we compare candidate geometries under a common protocol: same angle windows, same naturalness bounds, and the same normalized PMNS score used across all toys. The aggregate diagnostics are produced by `compute_flavour_geometry_comparison.py`. The score is defined as

$$S_{\rm PMNS} = \sqrt{\left(\frac{\theta_{12}-33^\circ}{10^\circ}\right)^2 + \left(\frac{\theta_{13}-8.6^\circ}{4^\circ}\right)^2 + \left(\frac{\theta_{23}-45^\circ}{7^\circ}\right)^2}, \tag{13}$$

and is combined with strict-yield fractions into a PMNS viability index in the summary JSON.

The main conclusion of this initial comparison is limited and explicit: within the current exploratory setup, topologically non-trivial geometries (torus ring and twisted torus) produce non-zero strict-window populations in large scans, while the tested rigid geometries (pyramidal, tetrahedral, spherical) remain at zero strict hits. This is the step that first singled out twisted-torus as the correct family direction. An intermediate phase-loop toy was also tested in the same spirit, but its dedicated flavour benchmark remained clearly below the twisted-torus and twisted-multi-ring results, so it is retained only as a superseded exploratory waypoint in the archived workspace.

Geometry	Representative score S_{PMNS}	Strict yield within natural scan	PMNS viability index
Pyramidal	2.40	— (single benchmark)	0.067
Tetrahedral apex/base	2.23	0/1500	0.107
Tetrahedral star	2.64	0/2701	0.013
Torus ring	0.78	59/4358	0.448
Spherical shell	2.69	0/4348	0.000
Twisted torus	0.52	82/4223	0.510

TABLE I. Initial cross-geometry exploratory comparison from `flavour_geometry_comparison_summary.json`.

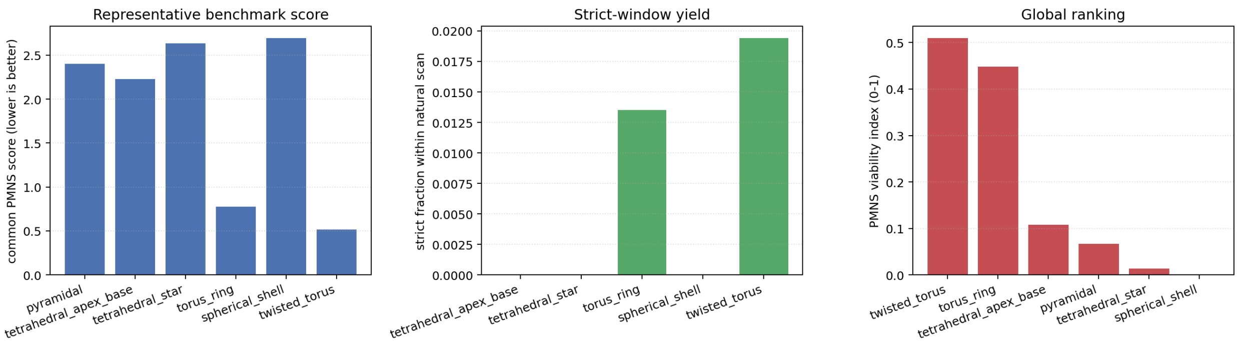

FIG. 6. Common-score and strict-yield comparison across explored cluster geometries.

We then refine this direction with a dedicated same-family comparison between twisted torus and twisted multi ring. The diagnostics are stored in `twisted_multi_ring_vs_twisted_torus_comparison.json`.

The dedicated twisted-multi-ring benchmark improves the twisted-torus score by roughly a factor of two and doubles the strict-yield fraction. In the logic of this paper, twisted torus remains the historical turning point, but twisted multi ring becomes the strongest current PMNS candidate.

VIII. SECTOR-COMPATIBILITY STRESS TESTS FOR TWISTED MULTI RING

We then test whether the same twisted-multi-ring geometry remains compatible with a hierarchical charged-lepton sector and with a weak-mixing quark-like toy sector. The diagnostics are generated by `compute_twisted_multi_ring_sector_com`

For charged leptons, perturbing the charged-sector controls around the best twisted-multi-ring point yields

$$f^{(\ell)}_{\mathrm{strict}} \simeq 0.87, \tag{14}$$

i.e. approximately 87% of tested points remain in the strict PMNS window. This is stronger than the historical twisted-torus robustness level ($\sim 70\%$) and therefore supports the same candidate shift seen in the PMNS benchmark itself. For the quark-like toy layer, the mean mixing angles stay small,

$$\langle\theta^q_{12}\rangle \simeq 0.35^\circ, \quad \langle\theta^q_{13}\rangle \simeq 1.2\times10^{-3\,\circ}, \quad \langle\theta^q_{23}\rangle \simeq 2.0\times10^{-3\,\circ}. \tag{15}$$

These tests support compatibility at the toy level, but do not constitute a complete SM flavour derivation.

IX. STATUS OF ASSUMPTIONS

- EFT-derived baseline definitions inherited from Papers I–IV.
- Phenomenological closures used in the CC-Φ_1 cluster pipeline.
- Exploratory geometry-to-flavour mapping for CC-Φ_2.
- Twisted-torus identified as the historical topology that opened the correct PMNS direction.
- Twisted-multi-ring selected as the strongest current exploratory flavour candidate, without changing the canonical EFT.

Candidate	S_{PMNS}	Strict fraction within natural scan
Twisted torus	0.519	0.0194
Twisted multi ring	0.235	0.0415

TABLE II. Dedicated PMNS-level comparison within the leading twisted-topology family.

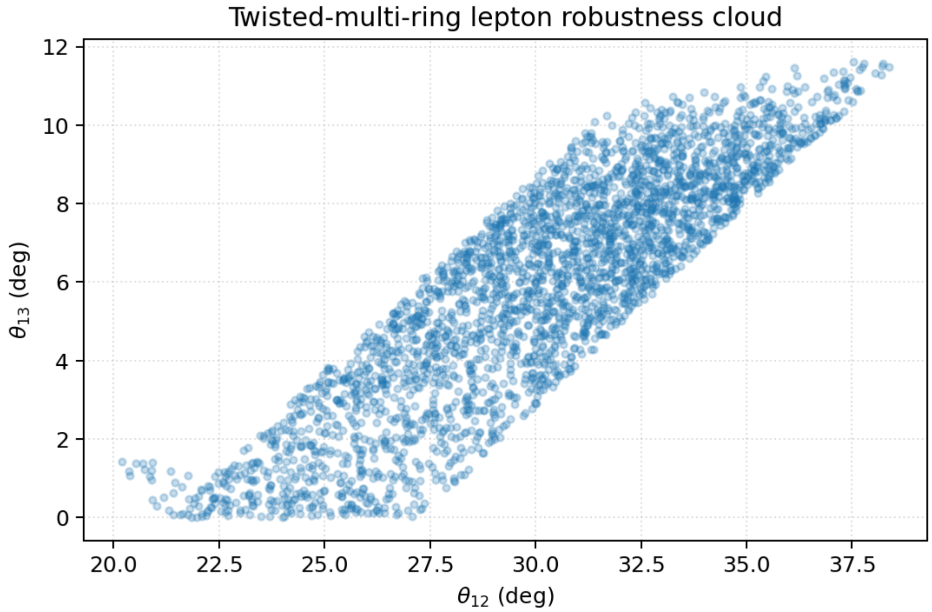

FIG. 7. Twisted-multi-ring lepton robustness cloud from `compute_twisted_multi_ring_sector_compatibility.py`.

X. CONCLUSION

Paper V is designed as a reproducible and conservative microstructure step in the TCV-ϕ series. Relative to earlier drafts, the flavour bridge is now organized as a geometry-selection problem: we test multiple cluster topologies under common criteria, identify twisted torus as the historical turning point, and then show that a refined twisted-multi-ring geometry becomes the strongest current exploratory candidate. This improves the naturalness narrative at the toy level while keeping all macro-level sectors (bounce, black holes, and cosmological fits) unchanged in this paper. A full micro-to-macro closure from this topological family is left for future work.

[1] C. Lecroq, "TCV-ϕ I: an effective cohesive–vacuum field theory and its ΛCDM-compatible background cosmology," preprint (2026), zenodo.19234640.
[2] C. Lecroq, "TCV-ϕ II: cohesive bounce, geometric reheating, and baseline baryogenesis/ULDM pipelines," preprint (2026), zenodo.19234953.
[3] C. Lecroq, "TCV-ϕ III: primordial perturbations across the cohesive bounce and CLASS/CAMB bridge diagnostics," preprint (2026), zenodo.192355333.
[4] C. Lecroq, "TCV-ϕ IV: cohesive non-singular black holes from static cores to slow rotation in the canonical EFT," preprint (2026), zenodo.19235896.
[5] I. Esteban *et al.*, "NuFIT 5.x global fit of neutrino oscillation parameters," online resource (accessed 2026), `https://www.nu-fit.org/`.

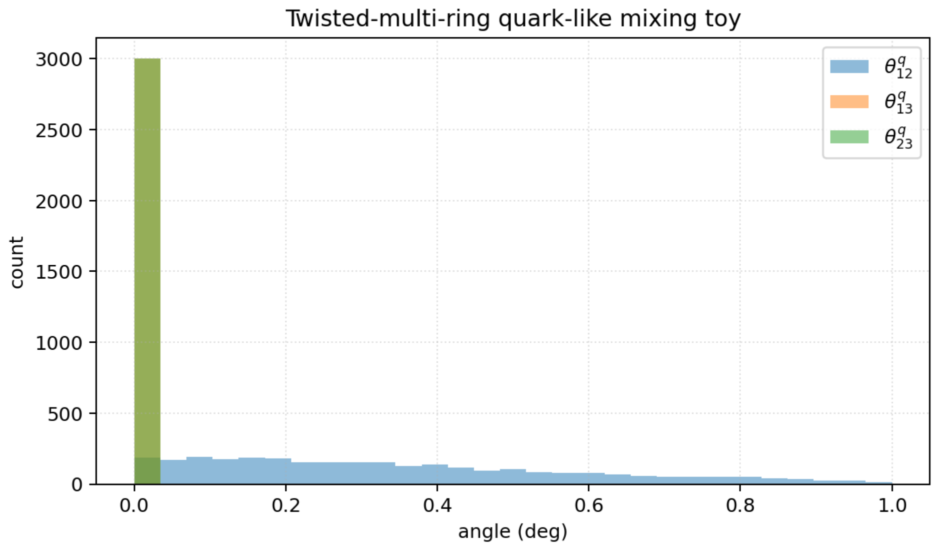

FIG. 8. Twisted-multi-ring quark-like toy mixing distribution. The angles remain small and hierarchical in this exploratory layer.

Block	Output artifact	Status label
CC-Φ_1 spectrum	`ccphi1_spectrum.json`	phenomenological toy
Curvature response	`ccphi1_curvature_scan.npz`	phenomenological scan
Coarse-graining	`ccphi1_coarse_grain.json`	toy proxy
Legacy pyramidal flavour toy	`ccphi2_flavour_toy.json`	exploratory legacy benchmark
Twisted-torus flavour toy	`twisted_torus_flavour_toy_summary.json`	exploratory historical benchmark
Phase-loop flavour toy	`phase_loop_flavour_toy_summary.json`	exploratory intermediate, superseded
Twisted-multi-ring flavour toy	`twisted_multi_ring_flavour_toy_summary.json`	exploratory current benchmark
Geometry comparison	`flavour_geometry_comparison_summary.json`	exploratory selection diagnostic
Twisted-family head-to-head	`twisted_multi_ring_vs_twisted_torus_comparison.json`	exploratory refinement diagnostic
Sector compatibility	`twisted_multi_ring_sector_compatibility_summary.json`	exploratory stress test

TABLE III. Paper-V script-backed outputs and status labels.